Team 12 Team Charter

Team Members	---------- ---------- ---------- PRINCE AKPOBASA **(**pakpobasa@gmail.com**)** ---------- ---------- ---------- ---------- ----------
Team Lead	Sam** ----------
Team Members Roles and Responsibilities	Team Members Roles and Responsibilities Sponsor Company – Client Individual Company Contacts – Client, Role(s) in Company ---------- - Team Lead, represents team to sponsor, via email and on calls, to minimize communication errors. ---------- - Project Manager, provides guidance and draws out insight from other team members, ensures that the project execution remains on track. PRINCE AKPOBASA (pakpobasa@gmail.com)- Project Scribe, responsible to taking meeting minutes and distributing notes/assignments. Can assist Team Lead in drafting emails and communication between sponsor and group. ---------- ---------- - Project Lead, responsible for holding the group accountable for meeting deadlines and ensures that the project deliverables are being met.

Mission, Vision Objectives & Core Values	*Mission (clear and concise language, providing actionable words that the group can stand for and accomplish):* 'To fulfil the needs of our sponsor through a tangible project plan and recommendations that they can execute in their company' *Vision Objectives (what does success look like?):* We want to work in a collaborative and positive team dynamic. It is the anchor point of any strategic plan.
	Core Values: Integrity, Accountability, Discipline, Respect, Innovation
Internal Checks, Balances, and Reviews	*A team charter is only as effective as the actions that are measured against it, so it's important for the team to spell out how internal checks and balances and reviews will be handled. What is expected of each sub-team and team member, and when? How often are full team check-ins, and what kind of topics should be covered? What are the goals of individual team members and how will they be measured? Consider the SMART goal acronym: Specific, Measurable, Achievable, Realistic, and Time-based.*

Operations: ● Assignments ● Meetings ● Communication Guidelines ● Status Updates ● Deadlines	*Assignments*: Reviewing the syllabus, creating the Project Plan, drafting the Project Report, delivering the Final Project. *Meetings:* Team will meet every Monday at 5pm via Microsoft Teams. Meeting with sponsor will take place every Tuesday at 11am via Skype. *Communication Guidelines:* Team Lead will represent team to sponsor; everyone is expected to participate and contribute and maintain collaboration (cc'ing entire team on emails, for example); main channel of communication among team will be email; emails among team members will be responded to within 24 hours, emails between Team Lead and Sponsor will be responded to within 48 hours; listen and respect each other's ideas, encourage conciseness *Status Updates:* Team will provide a weekly status update to client by Tuesday at 5pm. Individual contributions are due by Tuesday at 9am. *Deadlines:* Project Draft is due March 25th, Final project is due by April 18th.

Data visualization Week 1 Team 12

Prepared by:

PRINCE AKPOBASA

And 8 others

1. Dataset Overview & Documentation

Total Number of Rows and Columns:

- Rows: 7,543
- Columns: 80

Types of Variables:

- **Categorical:** Given_Name, Last_Name, College, Country, University, Intake, College_1st_Choice, Phone_Number, City, Region, Postal, Email_ID, Comments, and others (most variables are categorical).
- **Numerical:** A few columns, such as Reference_ID, Received_At, Date_of_Birth (but many numerical columns contain null or erroneous data).
- **Date:** Some columns like Start_Date, Created_At-3, Modified_At-3 are related to timestamps.

General Description of Dataset Themes:

The dataset appears to be related to educational records, potentially for students or applicants. It contains

information on names, contact details, addresses, educational choices, and timestamps for tracking. Some of the columns contain missing or null values, while others are specific to student-related data such as majors, college preferences, and application status.

2. Initial Observations or Notable Characteristics:

- **Missing Data:** Many columns have missing values, such as Major, Degree_Type, and

Date_of_Birth. These would require cleaning before further analysis.

- **Unusual Data Types:** Some columns, such as Phone_Number, contain a mix of string and numeric data.
- **Date Columns:** Several date-related columns may need conversion to proper date formats for further analysis.
- **Categorical Data Quality:** Some categorical columns like Country, City, and Region have some missing entries but appear to be important for the analysis.

Week 1 – Data Quality Report

1. Objective

The objective of this task was to evaluate the dataset quality by identifying:

- Missing values
- Duplicate records
- Columns with excessive missing data
- Outliers and unusual values
- Structural issues in the dataset

This assessment helps ensure the dataset is reliable before further analysis or modeling.

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2. Dataset Overview

- Total Records: **7543**
- Dataset contains a mix of:
    - Personal information fields ○ Application-related attributes
    - Contact history fields ○ System-generated columns

Many columns appear to be CRM/export based fields.

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3. Missing Value Analysis

Missing values were calculated for each column.

Key Findings

Columns with 100% Missing Values

A large number of columns contain **no data at all**, including:

- Major
- Degree_Type
- Counsler
- Citizenship
- Status
- Major_1st_Choice • Date_of_Birth
- Phone_Number_2
- Most_Recent_Released_Decision
- RIT_Email_Created
- Housing_Contract
- Is_Global_Grad
- Is_Admitted
- SEVIS_ID
- Official_University_email_address
- Application_Agency_Code
- Outstanding_Checklist_Items
- Prior_I-20_Outreach_Detail
- Prior_Non_I-20_Outreach_Detail
- Most_Recent_Contact
- Recent_GT_Form_Initiative

  Caller_Name_2

  Outcome_2

-
-

- Final_Result
- Escalation_Required
- SLU_Start_Comment
- SLU_Start_City
- I_901_Status
- Category
- Intake-2
- Status-2
- City_and_Branch
- Header
- Template
- Region-2

Insight:
These columns provide **zero analytical value** and should be **removed**.

Columns with High Missing Percentage (>50%)

- Street_2 → **61.5% missing**
- Remark → **91.06% missing**
- Remark_2 → **98.89% missing**

Impact:
Such columns are unreliable and may introduce bias.

Moderate Missing Columns

- Comments → 21.54%
- Admit_Date → 23.25%

-

- Region → 12.12%

  Phone_Number → 7.78%

These can be handled through:

- Imputation
- Business validation
- Optional removal

Low Missing Columns (<1%)

Examples:

- Given_Name
- College
- Country
- Intake
- Street_1
- City
- Email_ID

These are safe for analysis.

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4. Duplicate Record Analysis

Duplicate checks revealed:

- Reference_ID has **2543 duplicate occurrences**
  - Many columns show very high duplicate values **Interpretation**

This suggests:

Multiple records per student/application

- CRM export with repeated tracking entries

-

  Not necessarily data error

However:

- Duplicate **rows** should be removed
- Duplicate **IDs** must be validated with business logic

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5. Outlier / Unusual Value Analysis

Outliers detected mainly in:

- Recieved_At → **1015 unusual values** • Created_At → **1015 unusual values**
- Modified_At → **1015 unusual values**
- Campaign_Id → **122 unusual values**
- ID → **122 unusual values**

Meaning

Possible reasons:

- Incorrect timestamps
- System migration entries
- Bulk data uploads
- Abnormal campaign activity

These should be verified before removal.

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6. Structural Issues Observed Repeated

/ Versioned Columns

Columns like:

Reference_ID-2

Recieved_At-2

University-2

-
-

-
- University-3
- Created_At-2 / 3

Indicate:

- Multiple data sources
- Historical tracking
- Versioned CRM exports

These may require:

- Consolidation
- Feature selection

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7. Key Data Quality Problems Identified

Major issues in the dataset:

1. Large number of completely empty columns
2. High missing percentage fields
3. Duplicate IDs and repeated entries
4. Multiple redundant versioned columns
5. Timestamp anomalies
6. Inconsistent contact and campaign fields

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8. Recommendations

Before moving to analysis or ML:

Remove

All 100% missing columns

Columns with >90% missing

-
-

Clean

- Handle moderate missing columns
- Standardize timestamps
- Validate ID uniqueness

Transform

- Merge versioned columns
- Select only relevant business fields

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9. Conclusion

The data quality assessment revealed significant structural and completeness issues in the dataset. A large portion of columns contain missing or redundant information, which may affect analysis reliability.

Proper data cleaning steps such as removing empty columns, handling missing values, validating duplicates, and correcting anomalies are essential before proceeding to further stages like exploratory analysis or predictive modeling.

Improving data quality will ensure accurate insights and better decision-making.

Exploratory Data Analysis (EDA) Report

Dataset: DePaul Applicant & Outreach Data
Total Records: 7543 **Total Columns:** 80

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1. Objective

The purpose of this exploratory data analysis was to:

- Understand dataset structure
- Identify data quality issues
- Analyze distributions of key variables
- Detect missing data patterns

- Explore numerical and categorical behavior
- Generate early business insights for further modeling

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2. Dataset Overview

The dataset contains:

- Applicant identity information
- Contact details
- Academic preferences
- Outreach and campaign tracking fields

The dataset combines multiple operational sources, resulting in many columns with inconsistent coverage (**page 2 & page 11**)

Data Types

- Object columns: majority
- Numeric columns: very few
- Datetime fields present

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3. Data Cleaning and Preparation

Initial preprocessing included:

- Converting timestamp fields from epoch to datetime
- Standardizing missing values
- Removing whitespace duplicates
- Converting date columns to proper format

These steps improved readability and consistency before analysis (**page 4**)

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4. Missing Value Analysis

Missing values are a **major characteristic** of this dataset.

According to the missing value section (**page 11**):

- Many academic and decision-related fields are almost empty
- Identity and contact fields are relatively complete

Visual Evidence

- The **bar chart of top 25 missing columns** shows several columns with ~100% missing (**page 12**)
- The **missing heatmap** indicates sparsity patterns across records (**page 13**)

Interpretation

This suggests:

- Data merged from multiple systems
- Some outreach modules not used consistently
- Academic details not captured for all applicants

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5. Summary Statistics

Numeric variables show **limited coverage** and many contain null values (**page 5–8**)

Key observations:

- Only a small subset of columns has meaningful numeric data
- Many numeric columns represent IDs or timestamps
- Descriptive power is mainly in categorical features

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6. Categorical Distribution Analysis

Categorical plots show strong concentration in a few categories (**page 13**).

Country Distribution

From the bar chart (**page 14**):

- India dominates the dataset
- Followed by USA, Ghana, Nigeria, Pakistan

This indicates primary recruitment focus on Indian applicants.

Intake Distribution

From intake chart (**page 14**):

- Major programs include ○ Computer Science (MS Fall 2024)
  - ○ Business Analytics ○ Data Science

Shows technology-focused applicant pool.

Application Source

From chart (**page 15**):

- Most applications come from **Non Study Group**
- Smaller share from Study Group

This indicates direct application channels dominate.

College Preference

From chart (**page 15**):

- Jarvis College of Computing and Digital Media has highest preference
- Followed by Kellstadt School of Business

Shows demand for computing and business programs.

Outreach Outcomes

From chart (**page 16**):

- Most common outcome is **“ran”** and **Follow up**
- Some records indicate missing transcripts or pending actions

Indicates many applicants are still in early pipeline stages.

Slate Form Filled

From chart (**page 16**):

- Majority marked **Yes**
- Some still pending

Shows partial application completion.

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7. Numeric Distribution Analysis

Histograms on **pages 18–19** show:

- Attempts are highly concentrated around a single value
- Campaign_ID and ID are discrete clusters
- Reference_ID behaves like identifier, not analytical variable

Interpretation:

- Numeric fields mostly represent system identifiers
- Limited predictive usefulness

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8. Outlier Analysis

The boxplot for attempts (**page 20**) shows:

- Minimal variation

- Few extreme values

Outliers, if present, may indicate data entry or tracking anomalies.

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9. Correlation Analysis

The heatmap (**page 21**) shows:

- Weak meaningful correlations
- Some strong relationships caused by duplicated or timestamp columns

Reason:

Most numeric columns are identifiers, not behavioral metrics.

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10. Key Insights

Early insights summary (**page 21**) indicates:

- Dataset is strong for identity and geographic analysis
- Weak coverage for academic decision outcomes
- Country and intake are key segmentation variables
- Many columns may need to be dropped or treated as optional

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11. Business Understanding

From EDA:

- Majority applicants are from India
- Technology and business programs dominate
- Outreach process includes multiple follow-ups
- Application pipeline still incomplete for many records
- CRM system stores redundant tracking fields

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12. Conclusion

The exploratory analysis revealed that:

- The dataset is structurally complex with many missing fields
- Most informative variables are categorical rather than numeric
- Recruitment is heavily concentrated in specific countries and programs
- Many columns represent system tracking rather than analytical value
- Significant feature selection and cleaning is required before modeling

This EDA provides a strong foundation for:

- Data cleaning
- Feature engineering
- Applicant segmentation
- Predictive modeling in later stages.

In [69]:
```
# DePaul data, week 1 eda notebook
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

1. Dataset loading

The dataset was successfully loaded from the Excel file and contains applicant and outreach-related records for DePaul University.

In [ ]:
```
# 1) load dataset
file_path = "DePaul_Data.xlsx"
df = pd.read_excel('DePaul_Data.xlsx')
df
```

2. Dataset structure overview

The dataset contains a large number of columns, combining applicant identity, contact details, academic preferences, and outreach tracking fields. Several columns have very high missingness, suggesting some fields were not captured consistently or belong to a different pipeline stage.

In [55]:
```
# 2) quick dataset overview
print("shape, rows, columns")
print(df.shape)
print("\ncolumns")
print(df.columns.tolist())

print("\nmissing values percent, top 25")
missing_pct = (df.isna().mean() * 100).sort_values(ascending=False)
print(missing_pct.head(25))

print("\ndtypes")
print(df.dtypes.value_counts())
df
```

```
shape, rows, columns
(7543, 80)

columns
['Reference_ID', 'Given_Name', 'Last_Name', 'College', 'Major', 'Degree_Type', 'Coun
try', 'Recieved_At', 'Counsler', 'University', 'Citizenship', 'Status', 'Intake', 'C
ollege_1st_Choice', 'Major_1st_Choice', 'Phone_Number', 'Street_1', 'Street_2', 'Str
eet_3', 'City', 'Region', 'Postal', 'Email_ID', 'Date_of_Birth', 'Comments', 'Phone_
Number_2', 'Admit_Date', 'Most_Recent_Released_Decision', 'RIT_Email_Created', 'Hous
ing_Contract', 'Application_Source', 'Is_Global_Grad', 'Is_Admitted', 'SEVIS_ID', 'O
fficial_University_email_address', 'Application_Agency_Code', 'Created_At', 'Modifie
d_At', 'Reference_ID-2', 'Recieved_At-2', 'University-2', 'Outstanding_Checklist_Ite
ms', 'Prior_I-20_Outreach_Detail', 'Prior_Non_I-20_Outreach_Detail', 'Most_Recent_Co
ntact', 'Most_Recent_User_and_Date', 'Recent_GT_Form_Initiative', 'Date_of_Contact',
'Caller_Name', 'Outcome_1', 'Remark', 'Caller_Name_2', 'Date_of_Contact_2', 'Outcome
_2', 'Remark_2', 'Final_Result', 'Campaign_Id', 'Escalation_Required', 'SLU_Start_Co
mment', 'SLU_Start_City', 'I_901_Status', 'Slate_Form_Filled', 'Created_At-2', 'Modi
fied_At-2', 'ID', 'Name', 'Category', 'Attempts', 'Intake-2', 'University-3', 'Degre
e_Type-2', 'Status-2', 'City_and_Branch', 'Type', 'Header', 'Template', 'Start_Dat
e', 'Region-2', 'Created_At-3', 'Modified_At-3']

missing values percent, top 25
Most_Recent_Released_Decision        100.0
SLU_Start_Comment                    100.0
RIT_Email_Created                    100.0
Housing_Contract                     100.0
Is_Global_Grad                       100.0
Is_Admitted                          100.0
SEVIS_ID                             100.0
Official_University_email_address    100.0
Application_Agency_Code              100.0
I_901_Status                         100.0
SLU_Start_City                       100.0
Outstanding_Checklist_Items          100.0
Date_of_Birth                        100.0
Prior_I-20_Outreach_Detail           100.0
Prior_Non_I-20_Outreach_Detail       100.0
Most_Recent_Contact                  100.0
Most_Recent_User_and_Date            100.0
Recent_GT_Form_Initiative            100.0
Escalation_Required                  100.0
Final_Result                         100.0
Caller_Name_2                        100.0
Date_of_Contact_2                    100.0
Phone_Number_2                       100.0
Outcome_2                            100.0
City_and_Branch                      100.0
dtype: float64

dtypes
float64            46
object             29
int64               4
datetime64[ns]      1
Name: count, dtype: int64
```

Out[55]:

	Reference_ID	Given_Name	Last_Name	College	Major	Degree_Type	Country
0	45405320	Deeksha Reddy	Bhumireddy	INDIA - Osmania University - Bachelor's Degree	NaN	NaN	India
1	858032003	Pearl Ashok Kumar	Patel	INDIA - Manipal University Jaipur - Bachelor's...	NaN	NaN	India
2	902518555	Hamza	Javed	PAKISTAN - Shaheed Zulfikar Ali Bhutto Institu...	NaN	NaN	Pakistan
3	902518555	Hamza	Javed	PAKISTAN - Shaheed Zulfikar Ali Bhutto Institu...	NaN	NaN	Pakistan
4	218755608	Ronil Dhavalbhai	Thakkar	INDIA - Swarrnim Institute of Technology - Swa...	NaN	NaN	India
...	...	...	...	...	...	...	...
7538	644414774	Andy	Asiamah	GHANA - Kwame Nkrumah University of Science an...	NaN	NaN	Ghana
7539	514438657	Mines	Ghimire	INDIA - Sikkim manipal university - Master's D...	NaN	NaN	Canada
7540	514438657	Mines	Ghimire	INDIA - Sikkim manipal university - Master's D...	NaN	NaN	Canada
7541	514438657	Mines	Ghimire	INDIA - Sikkim	NaN	NaN	Canada

	Reference_ID	Given_Name	Last_Name	College	Major	Degree_Type	Country
				manipal university - Master's D...			
7542	514438657	Mines	Ghimire	INDIA - Sikkim manipal university - Master's D...	NaN	NaN	Canada

7543 rows × 80 columns

3. Basic cleaning and data type adjustments

Key timestamp fields appear to be stored as epoch values and were converted into readable datetime formats to support time-based exploration. Text fields were stripped to reduce duplication caused by spacing and to standardize missing value representations.

In [56]:
```
# 3) basic cleaning and type fixes
# convert unix-like timestamp columns if they look like milliseconds
epoch_cols = ["Recieved_At", "Created_At", "Modified_At", "Created_At-3", "Modified
for col in epoch_cols:
    if col in df.columns:
        if pd.api.types.is_numeric_dtype(df[col]):
            if df[col].dropna().astype(float).median() > 1e11:
                df[col + "_dt"] = pd.to_datetime(df[col], unit="ms", errors="coerce


# convert Start_Date if not already datetime
if "Start_Date" in df.columns:
    df["Start_Date"] = pd.to_datetime(df["Start_Date"], errors="coerce")

# convert selected date-like object columns to datetime
date_like_cols = ["Admit_Date", "Recieved_At-2", "Date_of_Contact"]
for col in date_like_cols:
    if col in df.columns:
        df[col + "_dt"] = pd.to_datetime(df[col], errors="coerce")

# strip text columns (helps reduce duplicates caused by spacing)
text_cols = df.select_dtypes(include=["object"]).columns
for col in text_cols:
    df[col] = df[col].astype(str).str.strip()
    df.loc[df[col].isin(["nan", "None", "NaT", ""]), col] = np.nan
```

4. Summary statistics

Numerical variables show limited coverage, with only a small subset of columns containing meaningful numeric values. Categorical fields provide stronger descriptive power, allowing us to summarize applicant distribution by country, intake, and application source.

In [57]:
```
# 4) summary statistics
print("\nsummary statistics, numeric")
print(df.describe(include=[np.number]).T)

print("\nsummary statistics, categorical (top freq)")
cat_cols = df.select_dtypes(include=["object"]).columns
for col in cat_cols[:12]:  # first 12 categorical columns
    print(f"\n{col}")
    print(df[col].value_counts(dropna=False).head(10))
```

summary statistics, numeric

	count	mean	std	\
Reference_ID	7543.0	4.731908e+08	2.984219e+08	
Major	0.0	NaN	NaN	
Degree_Type	0.0	NaN	NaN	
Recieved_At	7543.0	1.757495e+12	2.813731e+04	
Counsler	0.0	NaN	NaN	
Citizenship	0.0	NaN	NaN	
Status	0.0	NaN	NaN	
Major_1st_Choice	0.0	NaN	NaN	
Street_3	0.0	NaN	NaN	
Date_of_Birth	0.0	NaN	NaN	
Phone_Number_2	0.0	NaN	NaN	
Most_Recent_Released_Decision	0.0	NaN	NaN	
RIT_Email_Created	0.0	NaN	NaN	
Housing_Contract	0.0	NaN	NaN	
Is_Global_Grad	0.0	NaN	NaN	
Is_Admitted	0.0	NaN	NaN	
SEVIS_ID	0.0	NaN	NaN	
Official_University_email_address	0.0	NaN	NaN	
Application_Agency_Code	0.0	NaN	NaN	
Created_At	7543.0	1.757495e+12	2.813731e+04	
Modified_At	7543.0	1.757495e+12	2.813731e+04	
Reference_ID-2	4349.0	4.769586e+08	2.997553e+08	
Outstanding_Checklist_Items	0.0	NaN	NaN	
Prior_I-20_Outreach_Detail	0.0	NaN	NaN	
Prior_Non_I-20_Outreach_Detail	0.0	NaN	NaN	
Most_Recent_Contact	0.0	NaN	NaN	
Most_Recent_User_and_Date	0.0	NaN	NaN	
Recent_GT_Form_Initiative	0.0	NaN	NaN	
Caller_Name_2	0.0	NaN	NaN	
Date_of_Contact_2	0.0	NaN	NaN	
Outcome_2	0.0	NaN	NaN	
Final_Result	0.0	NaN	NaN	
Campaign_Id	4349.0	5.330050e+05	1.813321e+05	
Escalation_Required	0.0	NaN	NaN	
SLU_Start_Comment	0.0	NaN	NaN	
SLU_Start_City	0.0	NaN	NaN	
I_901_Status	0.0	NaN	NaN	
Created_At-2	4349.0	1.757494e+12	0.000000e+00	
Modified_At-2	4349.0	1.757494e+12	0.000000e+00	
ID	4349.0	5.330050e+05	1.813321e+05	
Category	0.0	NaN	NaN	
Attempts	4349.0	1.000000e+00	0.000000e+00	
Intake-2	0.0	NaN	NaN	
Status-2	0.0	NaN	NaN	
City_and_Branch	0.0	NaN	NaN	
Header	0.0	NaN	NaN	
Template	0.0	NaN	NaN	
Region-2	0.0	NaN	NaN	
Created_At-3	4349.0	1.757494e+12	0.000000e+00	
Modified_At-3	4349.0	1.757494e+12	0.000000e+00	

	min	25%	50%	\
Reference_ID	2.919250e+05	2.033014e+08	4.692245e+08	
Major	NaN	NaN	NaN	

Degree_Type	NaN	NaN	NaN
Recieved_At	1.757494e+12	1.757495e+12	1.757495e+12
Counsler	NaN	NaN	NaN
Citizenship	NaN	NaN	NaN
Status	NaN	NaN	NaN
Major_1st_Choice	NaN	NaN	NaN
Street_3	NaN	NaN	NaN
Date_of_Birth	NaN	NaN	NaN
Phone_Number_2	NaN	NaN	NaN
Most_Recent_Released_Decision	NaN	NaN	NaN
RIT_Email_Created	NaN	NaN	NaN
Housing_Contract	NaN	NaN	NaN
Is_Global_Grad	NaN	NaN	NaN
Is_Admitted	NaN	NaN	NaN
SEVIS_ID	NaN	NaN	NaN
Official_University_email_address	NaN	NaN	NaN
Application_Agency_Code	NaN	NaN	NaN
Created_At	1.757494e+12	1.757495e+12	1.757495e+12
Modified_At	1.757494e+12	1.757495e+12	1.757495e+12
Reference_ID-2	2.919250e+05	2.064377e+08	4.815995e+08
Outstanding_Checklist_Items	NaN	NaN	NaN
Prior_I-20_Outreach_Detail	NaN	NaN	NaN
Prior_Non_I-20_Outreach_Detail	NaN	NaN	NaN
Most_Recent_Contact	NaN	NaN	NaN
Most_Recent_User_and_Date	NaN	NaN	NaN
Recent_GT_Form_Initiative	NaN	NaN	NaN
Caller_Name_2	NaN	NaN	NaN
Date_of_Contact_2	NaN	NaN	NaN
Outcome_2	NaN	NaN	NaN
Final_Result	NaN	NaN	NaN
Campaign_Id	2.872300e+04	3.944880e+05	5.589930e+05
Escalation_Required	NaN	NaN	NaN
SLU_Start_Comment	NaN	NaN	NaN
SLU_Start_City	NaN	NaN	NaN
I_901_Status	NaN	NaN	NaN
Created_At-2	1.757494e+12	1.757494e+12	1.757494e+12
Modified_At-2	1.757494e+12	1.757494e+12	1.757494e+12
ID	2.872300e+04	3.944880e+05	5.589930e+05
Category	NaN	NaN	NaN
Attempts	1.000000e+00	1.000000e+00	1.000000e+00
Intake-2	NaN	NaN	NaN
Status-2	NaN	NaN	NaN
City_and_Branch	NaN	NaN	NaN
Header	NaN	NaN	NaN
Template	NaN	NaN	NaN
Region-2	NaN	NaN	NaN
Created_At-3	1.757494e+12	1.757494e+12	1.757494e+12
Modified_At-3	1.757494e+12	1.757494e+12	1.757494e+12

	75%	max
Reference_ID	7.362750e+08	9.995188e+08
Major	NaN	NaN
Degree_Type	NaN	NaN
Recieved_At	1.757495e+12	1.757495e+12
Counsler	NaN	NaN
Citizenship	NaN	NaN

Status	NaN	NaN
Major_1st_Choice	NaN	NaN
Street_3	NaN	NaN
Date_of_Birth	NaN	NaN
Phone_Number_2	NaN	NaN
Most_Recent_Released_Decision	NaN	NaN
RIT_Email_Created	NaN	NaN
Housing_Contract	NaN	NaN
Is_Global_Grad	NaN	NaN
Is_Admitted	NaN	NaN
SEVIS_ID	NaN	NaN
Official_University_email_address	NaN	NaN
Application_Agency_Code	NaN	NaN
Created_At	1.757495e+12	1.757495e+12
Modified_At	1.757495e+12	1.757495e+12
Reference_ID-2	7.516992e+08	9.990282e+08
Outstanding_Checklist_Items	NaN	NaN
Prior_I-20_Outreach_Detail	NaN	NaN
Prior_Non_I-20_Outreach_Detail	NaN	NaN
Most_Recent_Contact	NaN	NaN
Most_Recent_User_and_Date	NaN	NaN
Recent_GT_Form_Initiative	NaN	NaN
Caller_Name_2	NaN	NaN
Date_of_Contact_2	NaN	NaN
Outcome_2	NaN	NaN
Final_Result	NaN	NaN
Campaign_Id	6.010230e+05	8.259300e+05
Escalation_Required	NaN	NaN
SLU_Start_Comment	NaN	NaN
SLU_Start_City	NaN	NaN
I_901_Status	NaN	NaN
Created_At-2	1.757494e+12	1.757494e+12
Modified_At-2	1.757494e+12	1.757494e+12
ID	6.010230e+05	8.259300e+05
Category	NaN	NaN
Attempts	1.000000e+00	1.000000e+00
Intake-2	NaN	NaN
Status-2	NaN	NaN
City_and_Branch	NaN	NaN
Header	NaN	NaN
Template	NaN	NaN
Region-2	NaN	NaN
Created_At-3	1.757494e+12	1.757494e+12
Modified_At-3	1.757494e+12	1.757494e+12

summary statistics, categorical (top freq)

Given_Name

Given_Name

Fnu	101
Raj	53
Rahul	45
Syed Ali Mujtaba	33
Salman	31
Preksha Gowda	30
Sejal	30

```
Umar Javeed          27
Ishanshi             26
Dhruva Chaitanya     25
Name: count, dtype: int64

Last_Name
Last_Name
Patel       296
Mohammed    145
Shah         67
Gupta        55
Kundur       48
Khan         48
Lnu          47
Parmar       44
Syed         41
Singh        37
Name: count, dtype: int64

College
College
INDIA - Osmania University - Bachelor's Degree
269
INDIA - Gujarat Technological University - Bachelor's Degree
74
INDIA - University of Mumbai - Bachelor's Degree
71
NaN
71
INDIA - Visvesvaraya Technological University - Bachelor's Degree
62
INDIA - Anna University - Bachelor's Degree
60
INDIA - Jawaharlal Nehru Technological University Hyderabad - Bachelor's Degree
56
INDIA - Maharashtra State Board of Technical Education - Other\nINDIA - K J College
Of Engineering Management and Research - Bachelor's Degree     48
INDIA - Savitribai Phule Pune University - Bachelor's Degree
47
GHANA - Kwame Nkrumah University of Science and Technolog - Bachelor's Degree
40
Name: count, dtype: int64

Country
Country
India            4617
United states     764
Ghana             440
Nigeria           430
Pakistan          273
China             125
Nepal              81
Iran               75
Canada             69
Kenya              62
Name: count, dtype: int64
```

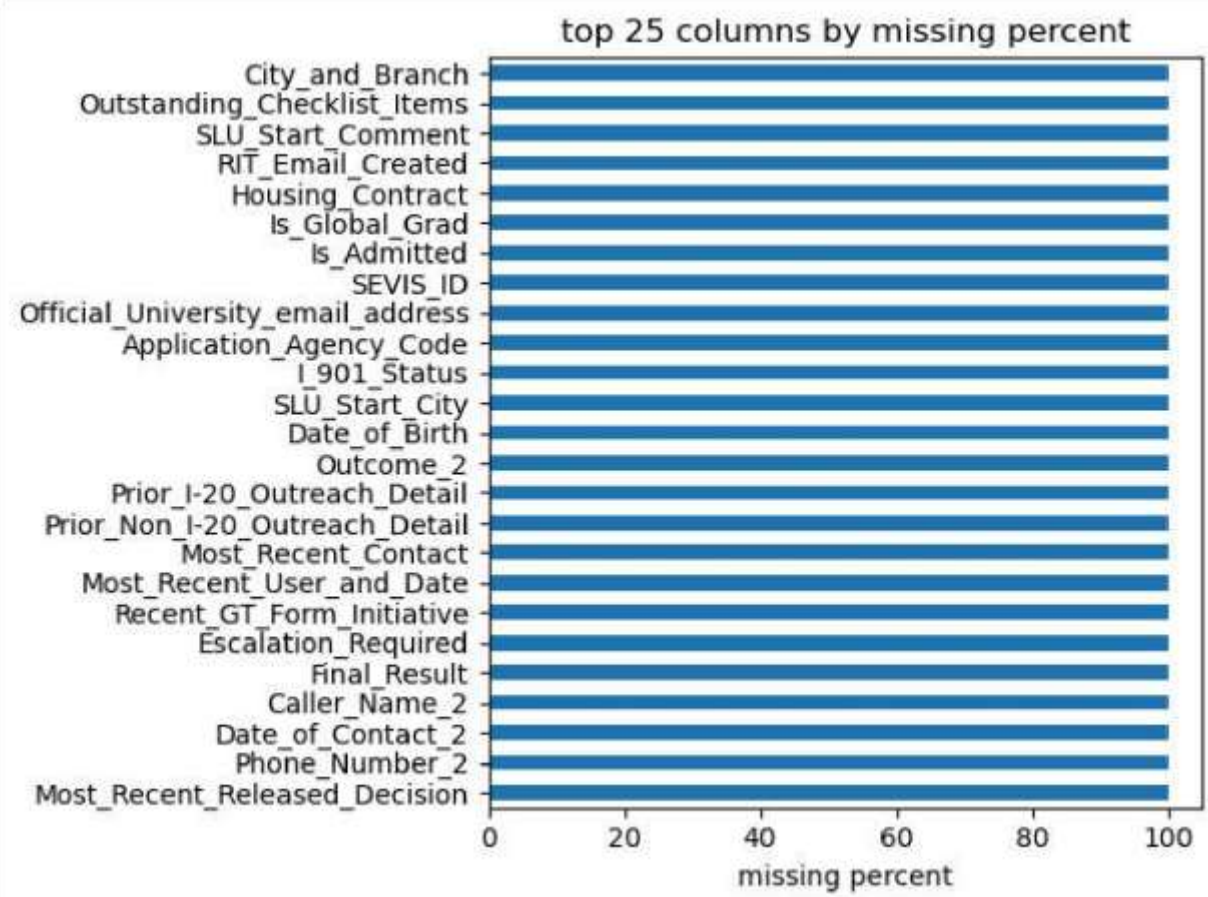
```
University
University
DePaul University    7543
Name: count, dtype: int64

Intake
Intake
Computer Science - MS: 2024 Fall (September '24)              504
Business Analytics - MS: 2024 Fall (September '24)            419
Data Science - MS: 2024 Fall (September '24)                  254
Cybersecurity - MS: 2024 Fall (September '24)                 181
Business Analytics - MS: 2025 Fall (September '25)            160
Artificial Intelligence - MS: 2024 Fall (September '24)       160
Health Informatics - MS: 2024 Fall (September '24)            155
Business Analytics - MS: 2025 Winter (January '25)            144
Business Analytics - MS: 2024 Winter (January '24)            143
Human-Computer Interaction - MS: 2024 Fall (September '24)    140
Name: count, dtype: int64

College_1st_Choice
College_1st_Choice
Jarvis College of Computing and Digital Media                   3831
Kellstadt Graduate School Of Business                           2523
College of Science and Health                                    269
College of Education                                             208
College of Liberal Arts and Social Sciences                      207
College of Communication                                          87
School of Psychology                                              79
School of Public Service                                          77
Jarvis College of Computing and Digital Media (Fully Online)      72
School of Music                                                   35
Name: count, dtype: int64

Phone_Number
Phone_Number
NaN                 587
920000000000         43
919000000000         39
+91 84290 90019      33
+91 96764 94150      31
+91 80088 47896      30
+91 97899 88124      30
+91 91217 94780      27
+1 872-338-6722      26
+91 7019941170       25
Name: count, dtype: int64

Street_1
Street_1
Omkarpark Phase-2,Rajmudra Society, Dhankawadi        48
45F Mohalla Kundigarrh Post Sadar                     33
H.No:-9-4-136/146, Jamalikunta, Tolichowki            31
6-1-306/B56 Old Cib Quarters Khairatabad              30
Annapurneshwari Krupa, Near Royal Apollo              30
10-5-4/A Masab Tank                                   27
```

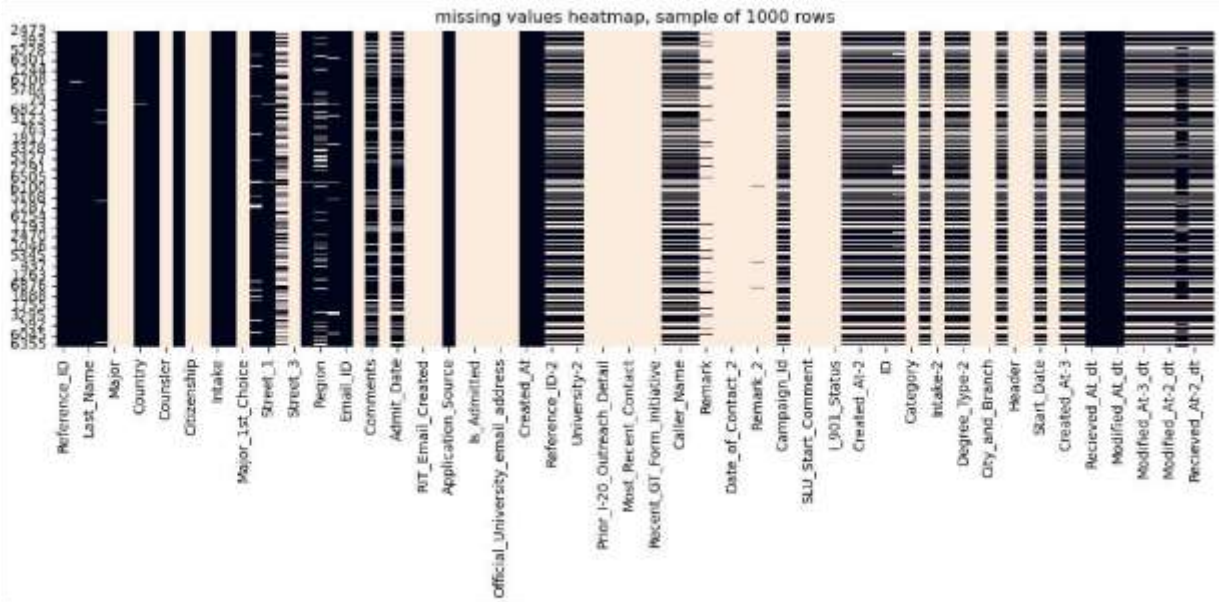
```
13 Shreenath Soc, Usmanpura Ashram Road, Ahmedabad    26
No 57,14th Cross, Mts Layout, Kengeri                 25
60 Sainik Vihar, Pitampura                            23
74 Prakruti Banglows Sterling                         23
Name: count, dtype: int64

Street_2
Street_2
NaN                                          4639
School Hemavathi Nagara                        30
Secunderabad                                   30
Musheerabad                                    26
Satellite Town                                 25
City Sector F  Bhopal Ahemdabad                23
Silent Zone, Opp. Airport Road, Dumas          22
Sri Ram Park Colony, Road No.8,                21
Lbs Marg                                       20
Flat 102, Mumtaz College Road,                 20
Name: count, dtype: int64

City
City
Hyderabad    975
Chicago      268
Ahmedabad    223
Chennai      206
Mumbai       204
Pune         164
Accra        158
Bengaluru    134
Karachi      108
Surat         96
Name: count, dtype: int64

Region
Region
Telangana         1239
NaN                914
Maharashtra        645
IL                 524
Gujarat            486
Andhra Pradesh     406
Karnataka          338
Tamil Nadu         288
Greater Accra      208
Lagos              149
Name: count, dtype: int64
```

5. Missing values and data quality

Missing values are a major characteristic of this dataset. Many academic and status-related fields are nearly empty, while core identity and contact fields are more complete. This suggests the dataset may be merged from multiple sources, where only some records include outreach and campaign details.

In [58]:
```
# 5) missingness visualization
# missingness bar, top 25 missing columns
plt.figure()
missing_pct.head(25).sort_values().plot(kind="barh")
plt.title("top 25 columns by missing percent")
plt.xlabel("missing percent")
plt.tight_layout()
plt.show()

# missingness heatmap (sampled for readability)
sample_df = df.sample(min(len(df), 1000), random_state=42)

plt.figure(figsize=(12, 6))
try:
    import seaborn as sns
    sns.heatmap(sample_df.isna(), cbar=False)
    plt.title("missing values heatmap, sample of 1000 rows")
except Exception:
    plt.imshow(sample_df.isna(), aspect="auto")
    plt.title("missing values heatmap, sample of 1000 rows")
    plt.xlabel("columns")
    plt.ylabel("rows")

plt.tight_layout()
plt.show()
```

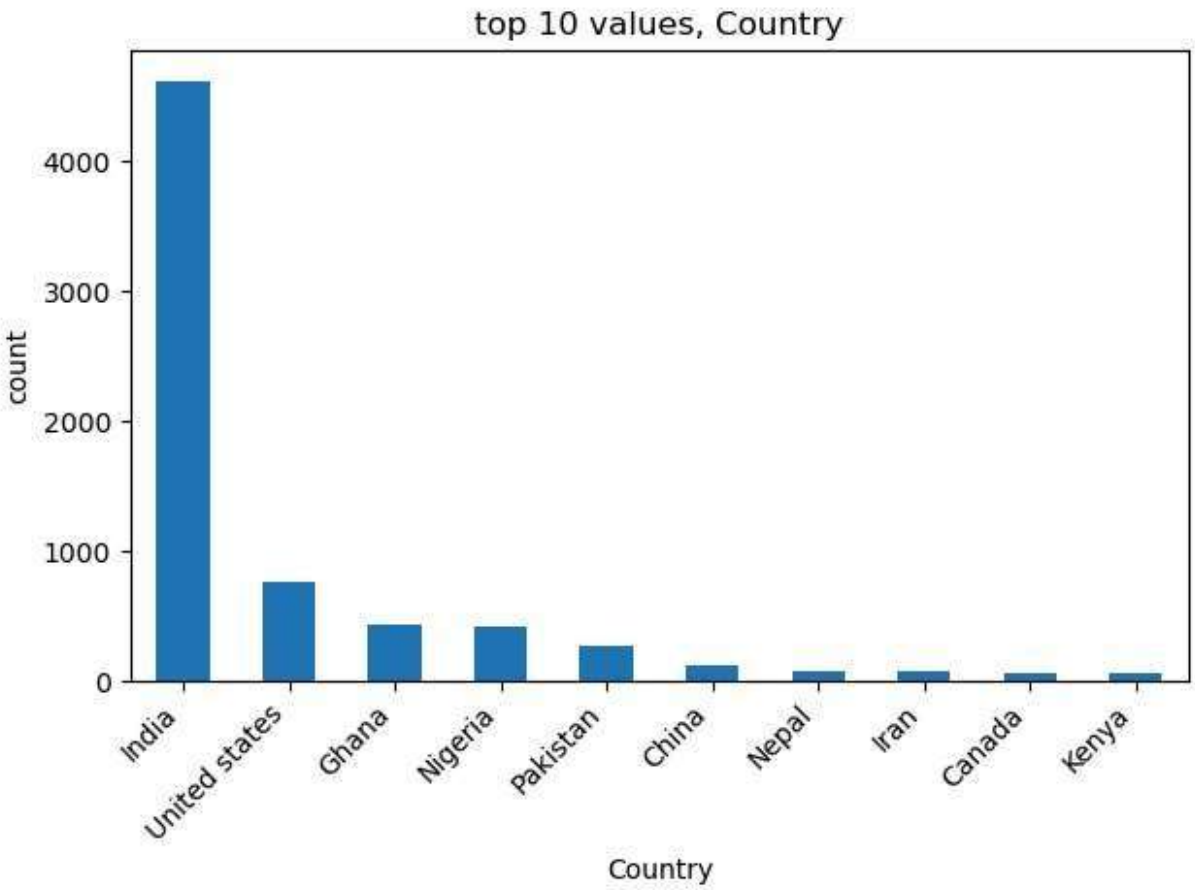


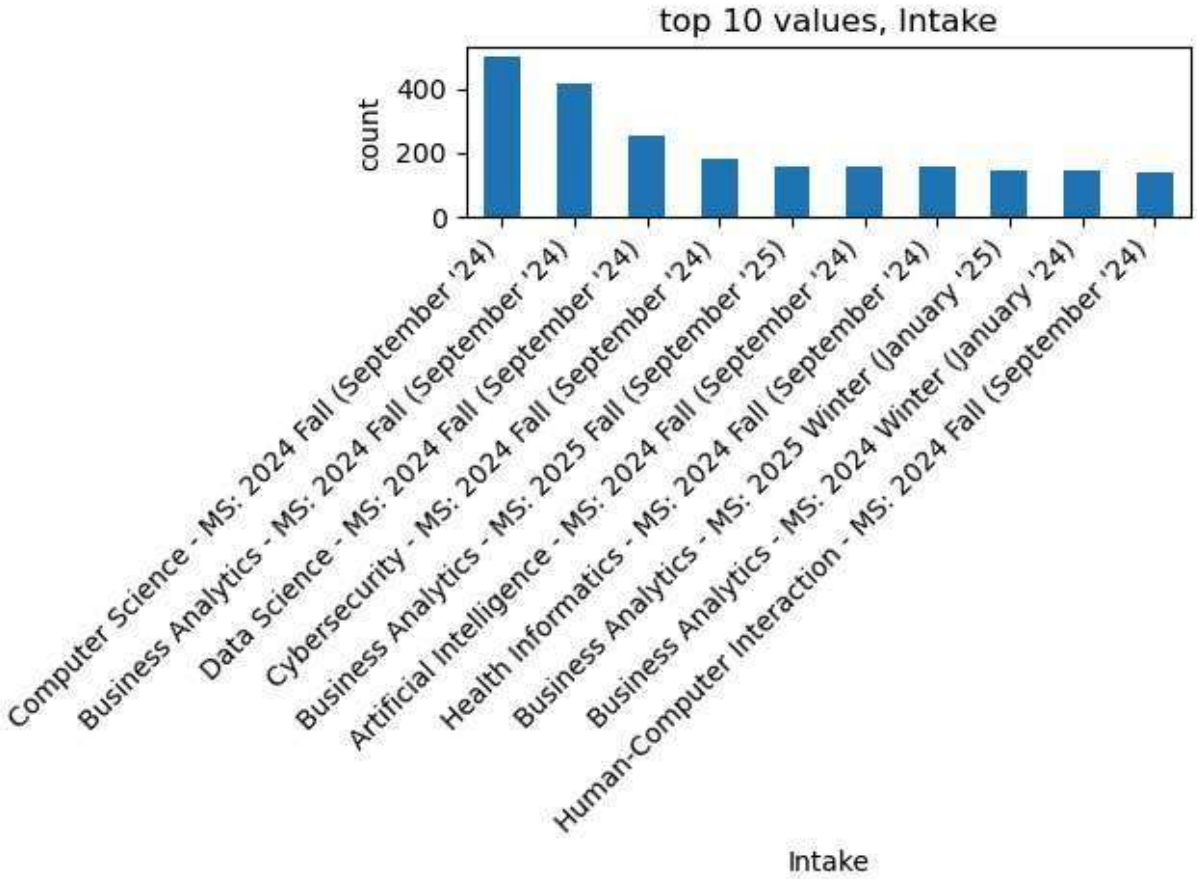


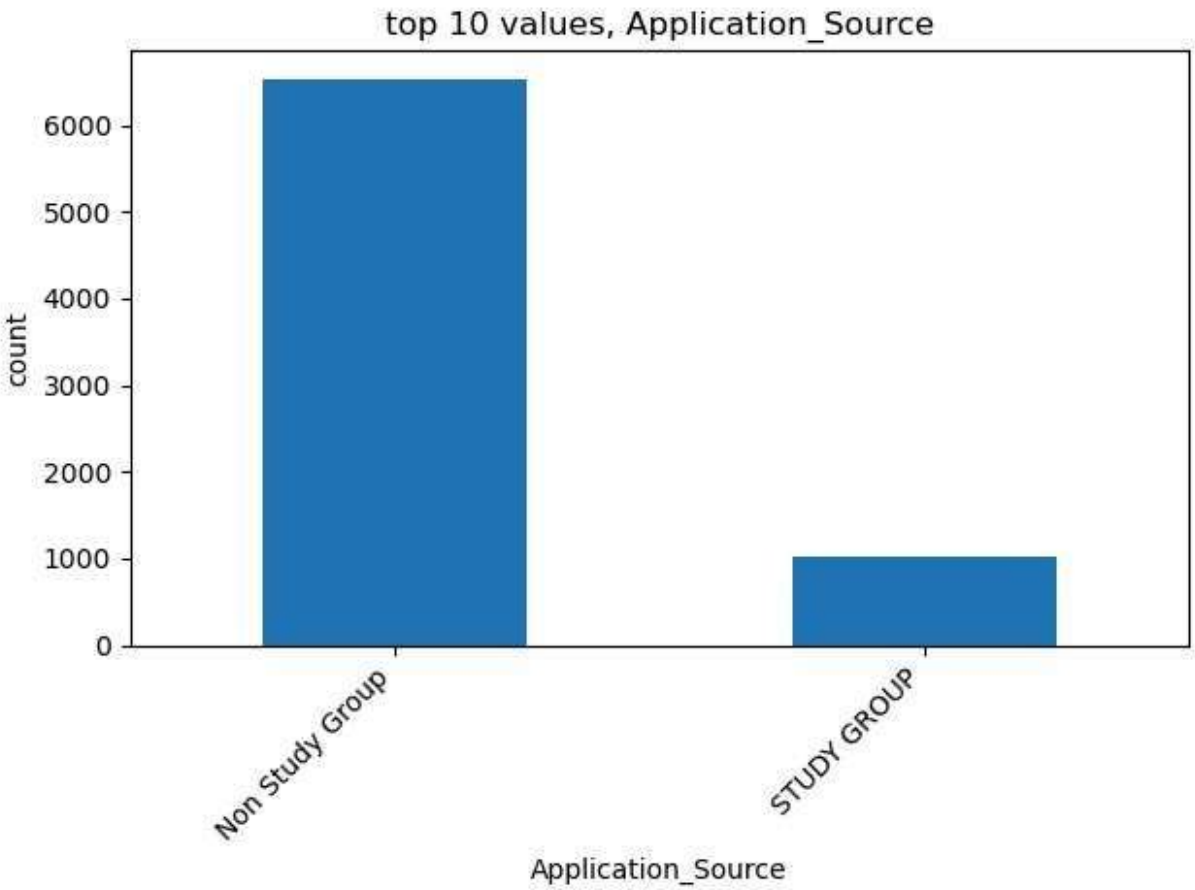
6. Categorical, distributions

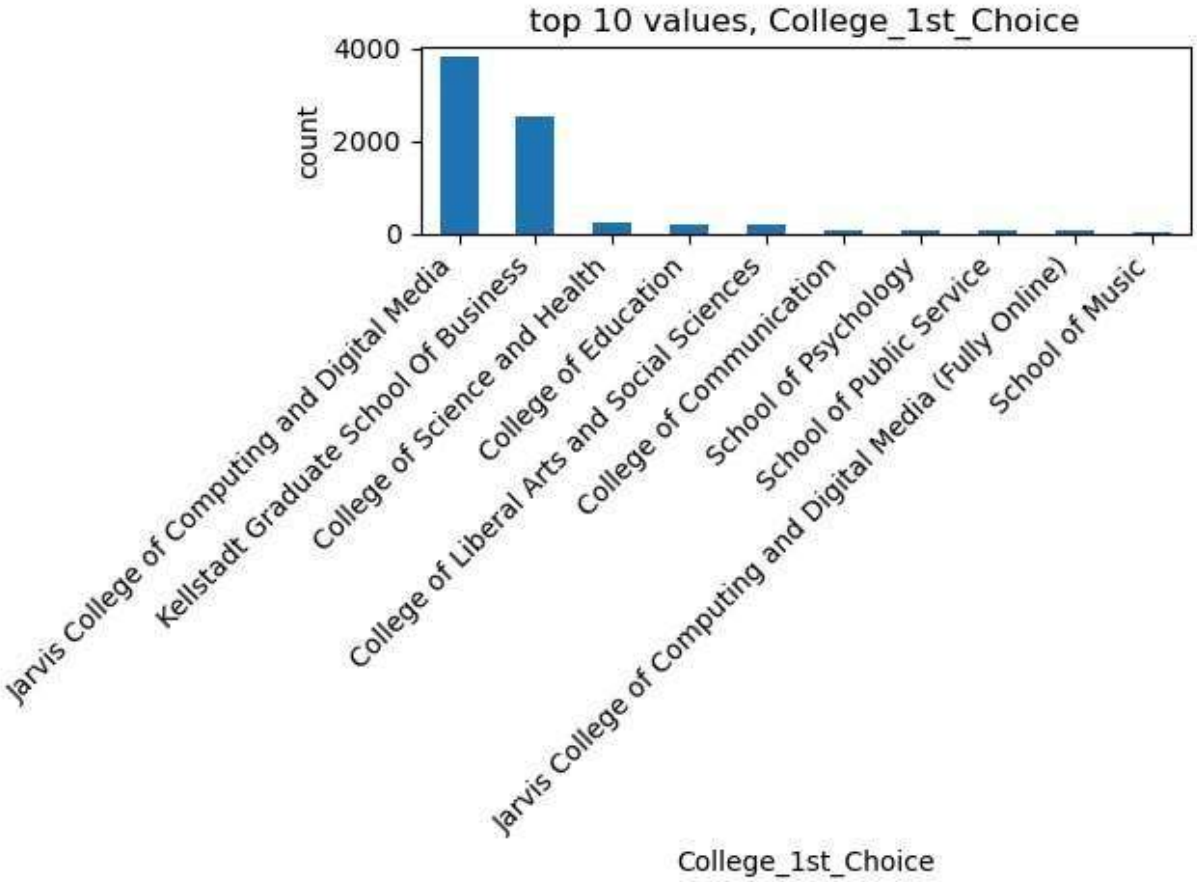
The categorical plots reveal concentration patterns where a few categories dominate the dataset, while many others appear infrequently. This is useful for identifying the main applicant segments and prioritizing the most informative categories for future analysis.

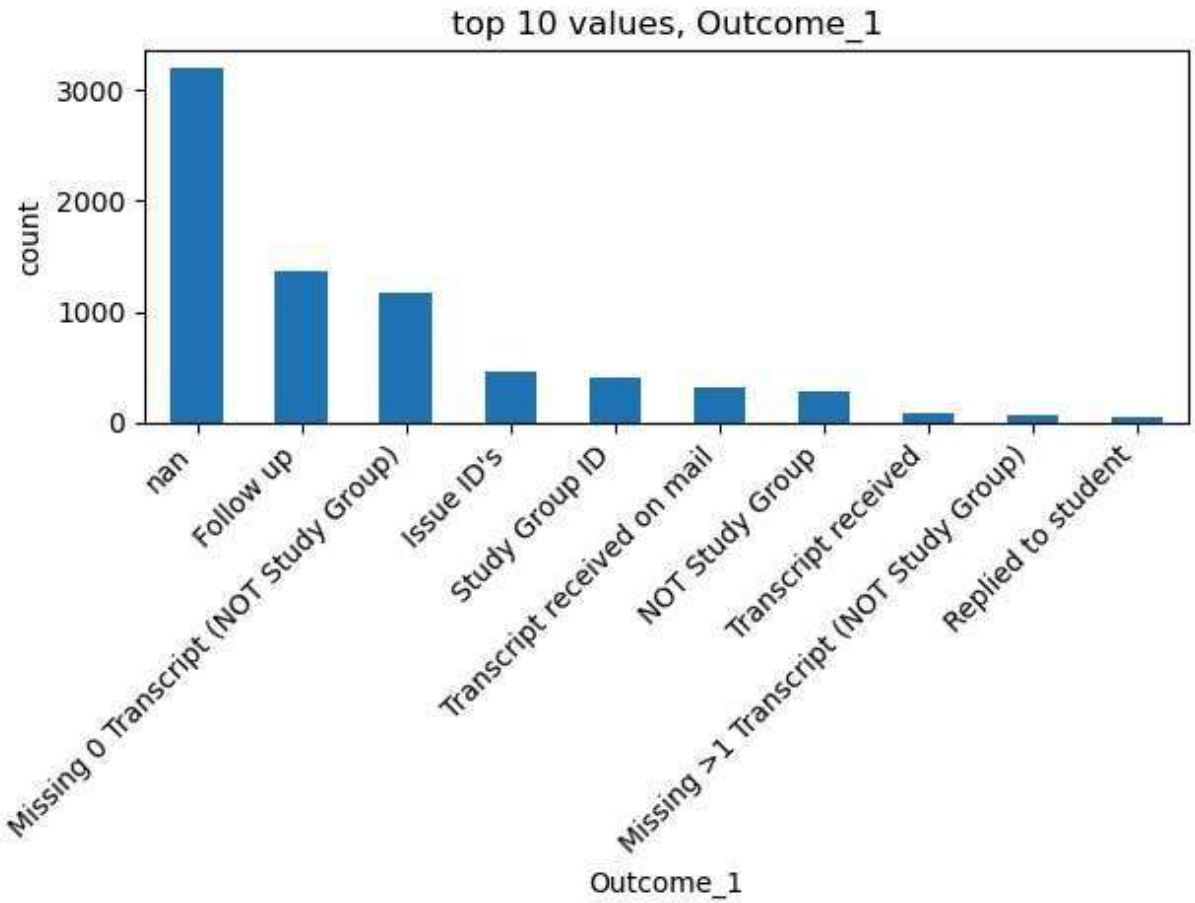
In [59]:
```
# 6) key categorical plots
def plot_top_categories(col, top_n=10):
    if col not in df.columns:
        print(f"skipping, column not found, {col}")
        return
    vc = df[col].value_counts(dropna=False).head(top_n)
    plt.figure()
    vc.plot(kind="bar")
    plt.title(f"top {top_n} values, {col}")
    plt.ylabel("count")
    plt.xticks(rotation=45, ha="right")
    plt.tight_layout()
    plt.show()

for col in ["Country", "Intake", "Application_Source", "College_1st_Choice", "Outco
    plot_top_categories(col, top_n=10)
```

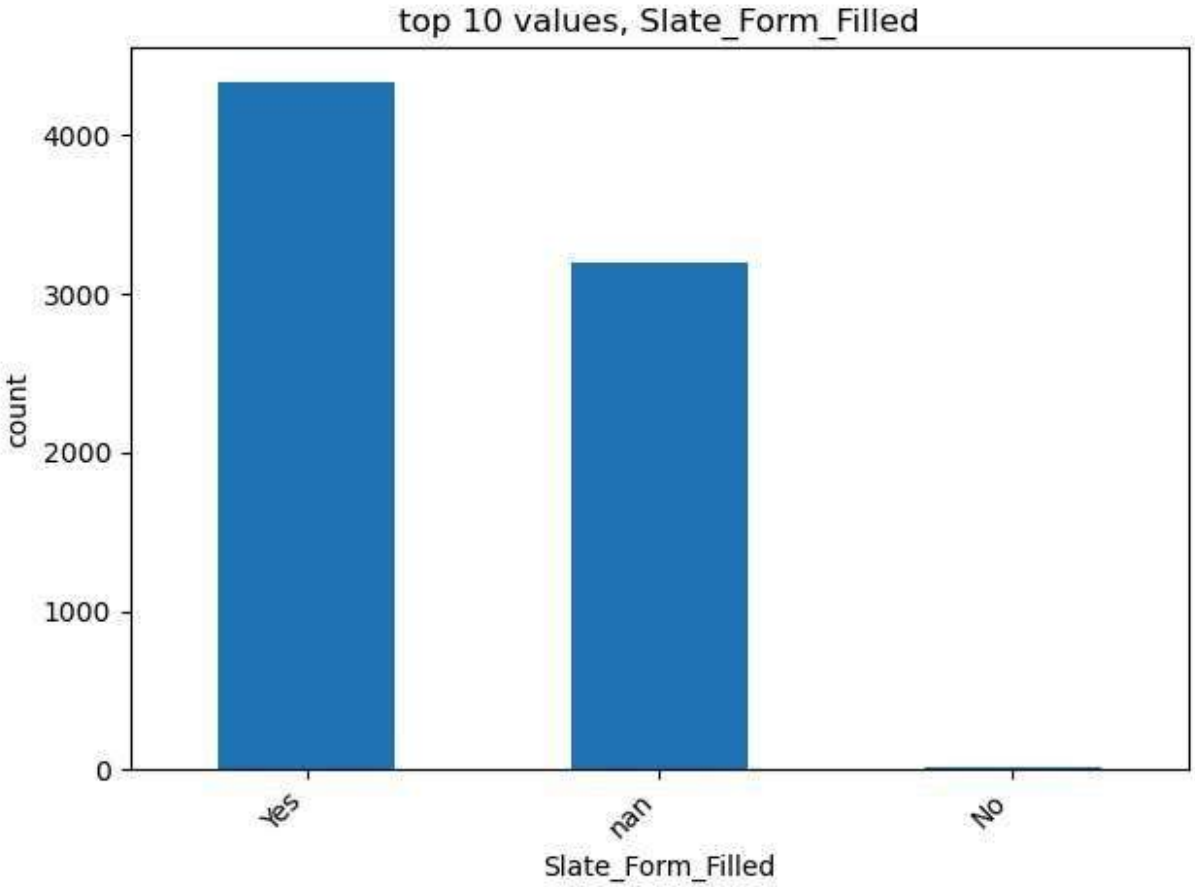

top 10 values, Country
count
4000
3000
2000
1000
0
India
United states
Ghana
Nigeria
Pakistan
China
Nepal
Iran
Canada
Kenya
Country


top 10 values, Intake
count
400
200
0
Computer Science - MS: 2024 Fall (September '24)
Business Analytics - MS: 2024 Fall (September '24)
Data Science - MS: 2024 Fall (September '24)
Cybersecurity - MS: 2024 Fall (September '25)
Business Analytics - MS: 2025 Fall (September '24)
Artificial Intelligence - MS: 2024 Fall (September '24)
Health Informatics - MS: 2024 Fall (September '24)
Business Analytics - MS: 2025 Winter (January '25)
Business Analytics - MS: 2024 Winter (January '24)
Human-Computer Interaction - MS: 2024 Fall (September '24)
Intake


top 10 values, Application_Source
count
6000
5000
4000
3000
2000
1000
0
Non Study Group
STUDY GROUP
Application_Source


top 10 values, College_1st_Choice
count
4000
2000
0
Jarvis College of Computing and Digital Media
Kellstadt Graduate School Of Business
College of Science and Health
College of Education
College of Liberal Arts and Social Sciences
College of Communication
School of Psychology
School of Public Service
Jarvis College of Computing and Digital Media (Fully Online)
School of Music
College_1st_Choice

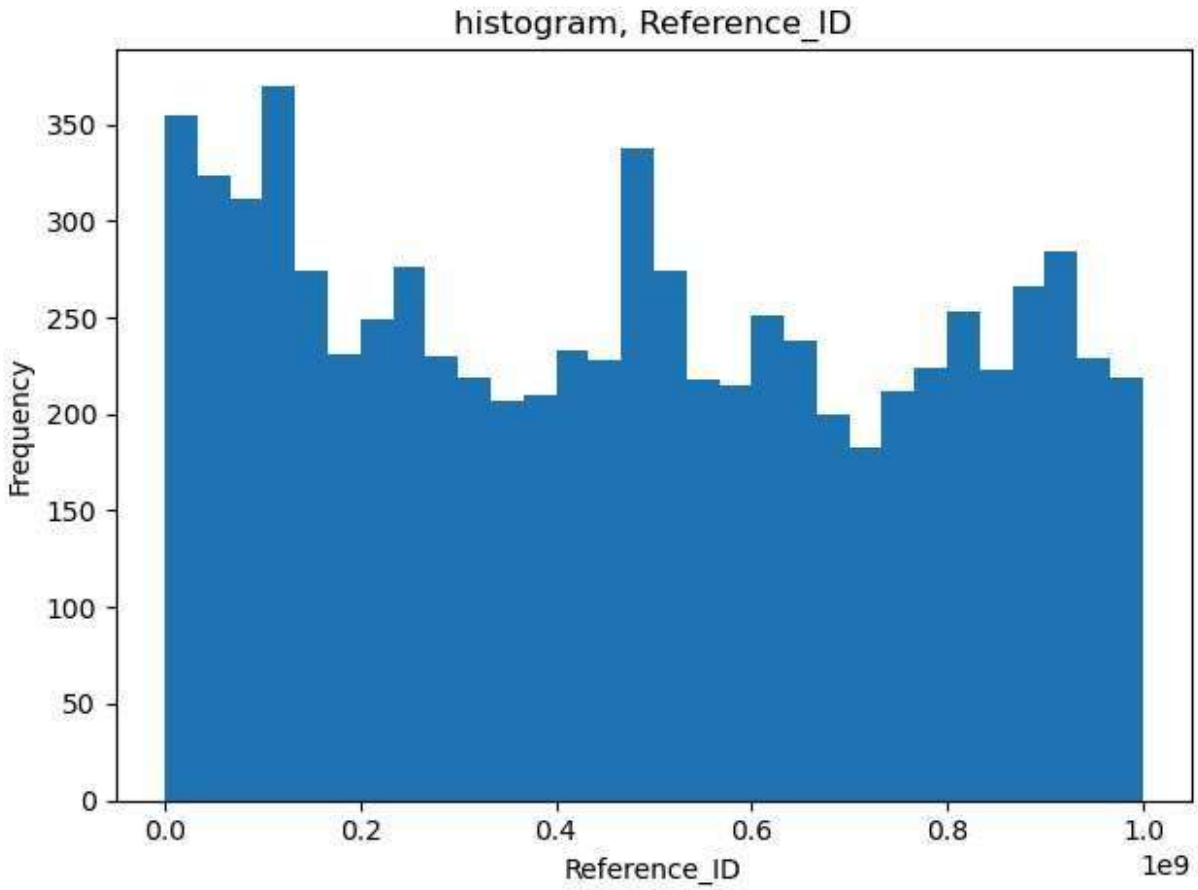

top 10 values, Outcome_1
count
3000
2000
1000
0
nan
Follow up
Missing 0 Transcript (NOT Study Group)
Issue ID's
Study Group ID
Transcript received on mail
NOT Study Group
Transcript received
Missing >1 Transcript (NOT Study Group)
Replied to student
Outcome_1

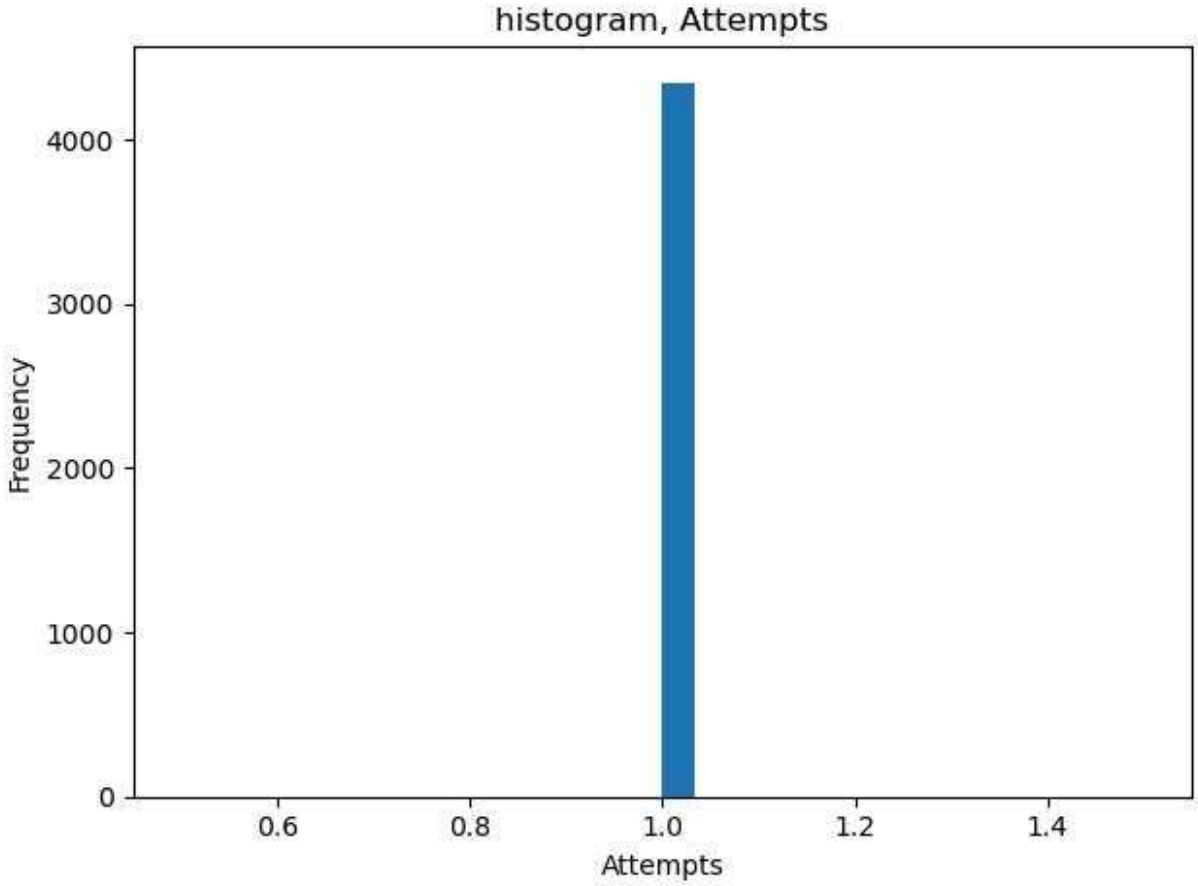

top 10 values, Slate_Form_Filled
count
4000
3000
2000
1000
0
Yes
nan
No
Slate_Form_Filled

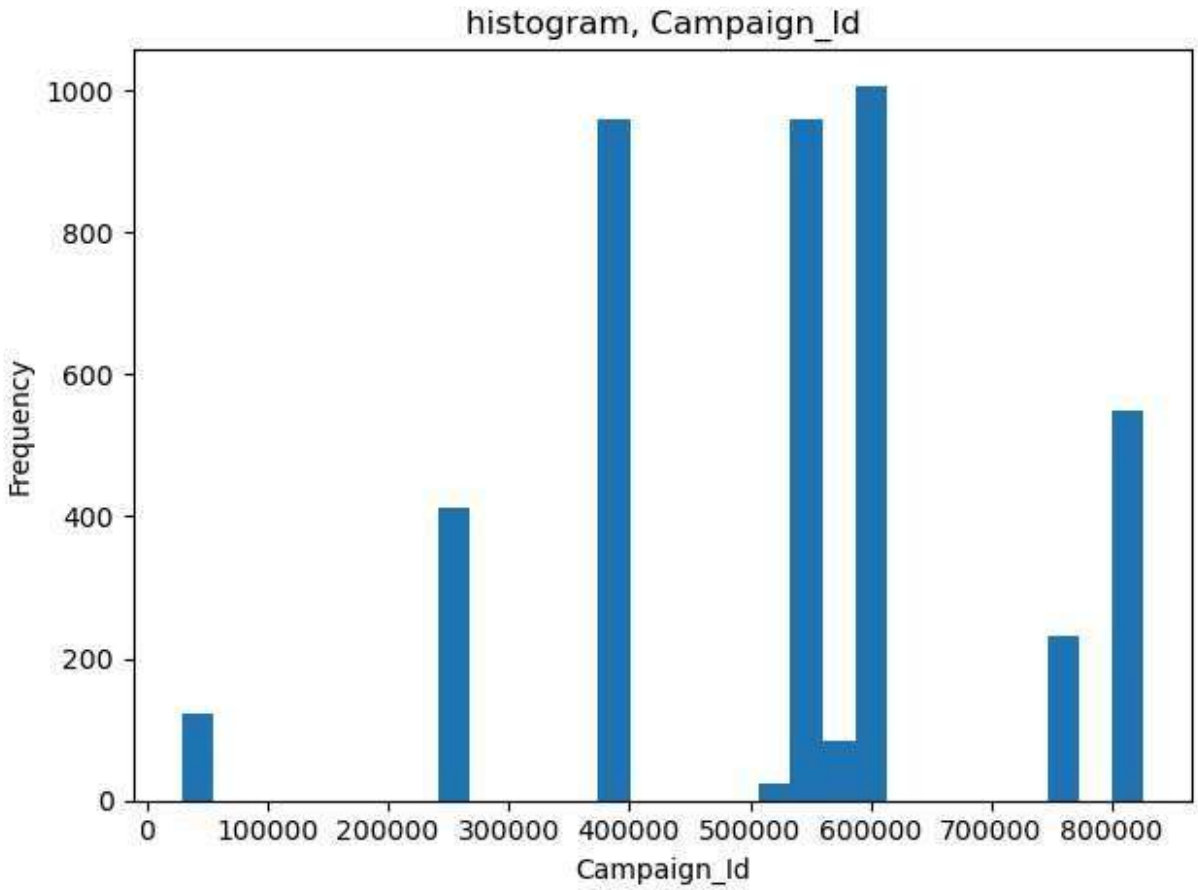
7. Numeric distributions

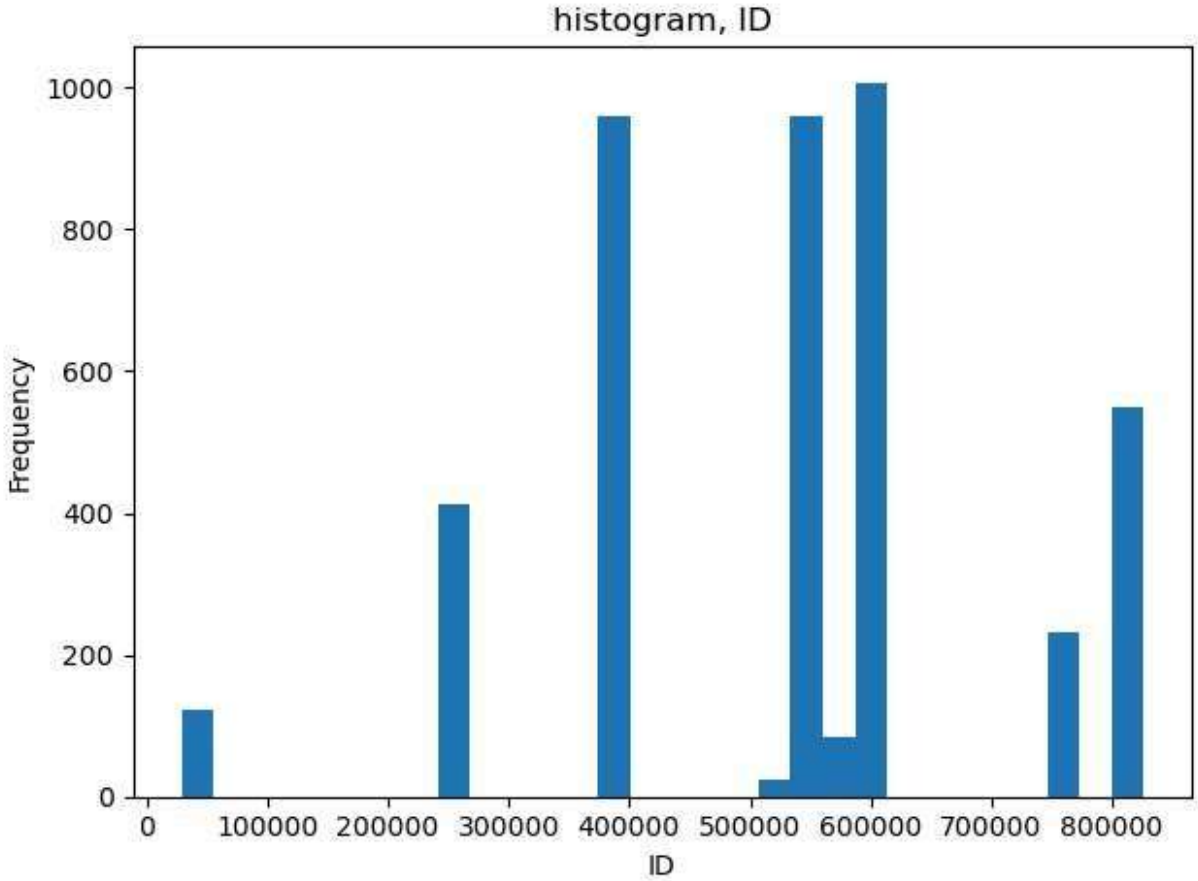
Show whether values are balanced or skewed. For example, attempts typically show a right-skewed pattern in outreach datasets, where most applicants require few attempts, while a minority require repeated follow-up. Outliers, if present, should be reviewed to confirm whether they reflect genuine cases or data entry issues.

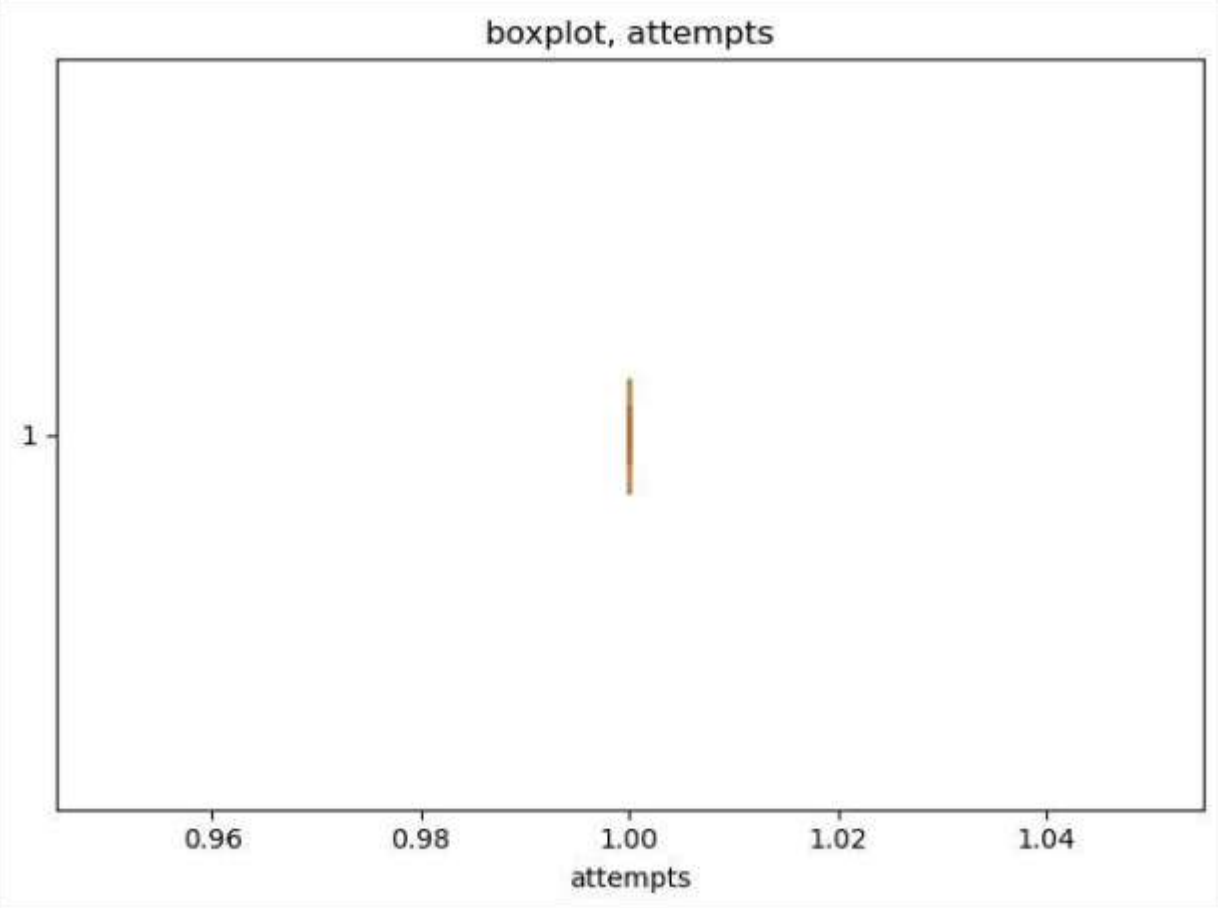
In [67]:
```
# 7) numeric distributions
def plot_hist(col, bins=30):
    if col not in df.columns:
        print(f"skipping, column not found, {col}")
        return
    if not pd.api.types.is_numeric_dtype(df[col]):
        print(f"skipping, not numeric, {col}")
        return
    plt.figure()
    df[col].dropna().plot(kind="hist", bins=bins)
    plt.title(f"histogram, {col}")
    plt.xlabel(col)
    plt.tight_layout()
    plt.show()

for col in ["Reference_ID", "Attempts", "Campaign_Id", "ID"]:
    plot_hist(col)

# boxplot for attempts (if available)
if "Attempts" in df.columns and pd.api.types.is_numeric_dtype(df["Attempts"]):
    plt.figure()
    plt.boxplot(df["Attempts"].dropna(), vert=False)
    plt.title("boxplot, attempts")
    plt.xlabel("attempts")
    plt.tight_layout()
    plt.show()
```


histogram, Reference_ID
Frequency
350
300
250
200
150
100
50
0
0.0
0.2
0.4
0.6
0.8
1.0
1e9
Reference_ID


histogram, Attempts
Frequency
4000
3000
2000
1000
0
0.6
0.8
1.0
1.2
1.4
Attempts


histogram, Campaign_Id
Frequency
1000
800
600
400
200
0
0
100000
200000
300000
400000
500000
600000
700000
800000
Campaign_Id


histogram, ID
Frequency
1000
800
600
400
200
0
0
100000
200000
300000
400000
500000
600000
700000
800000
ID

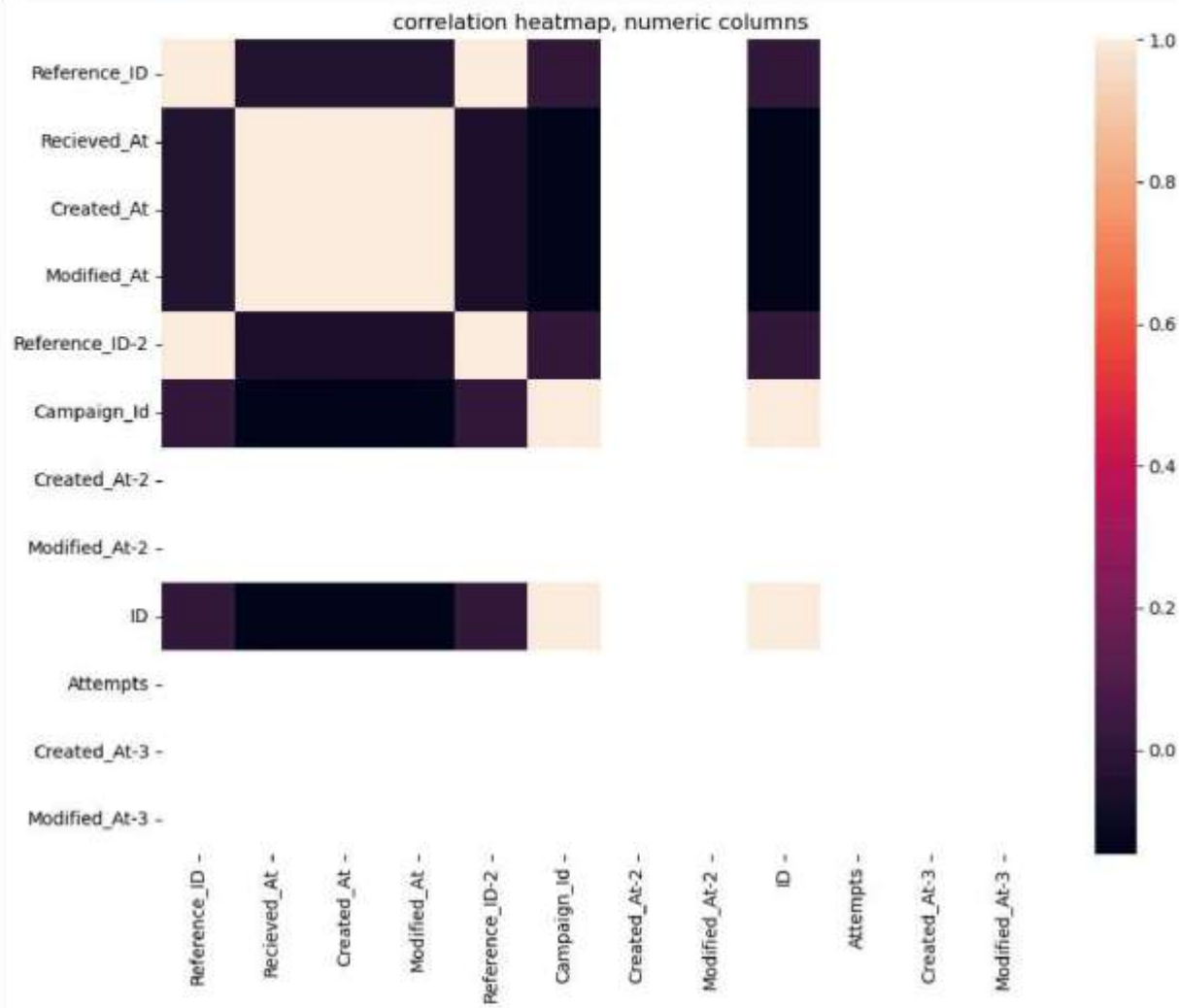


8. Correlations

The correlation heatmap focuses only on available numeric columns. At this early stage, correlations are expected to be weak or limited because many important variables are categorical and several numeric fields may represent identifiers rather than measurable quantities. Any strong correlation observed should be double-checked to ensure it is not driven by duplicated columns or timestamp encoding.

In [66]:
```
# 8) correlations + heatmap

num_df = df.select_dtypes(include=[np.number]).copy()

# drop columns that are fully missing
num_df = num_df.dropna(axis=1, how="all")

corr = num_df.corr(numeric_only=True)

plt.figure(figsize=(10, 8))
try:
    import seaborn as sns
    sns.heatmap(corr, annot=False)
except ImportError:
    plt.imshow(corr.values, aspect="auto")
    plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
    plt.yticks(range(len(corr.index)), corr.index)
```

```
plt.title("correlation heatmap, numeric columns")
plt.tight_layout()
plt.show()
```



9. Early insights summary

Early exploration suggests the dataset has strong coverage for applicant identity and geographic information, but incomplete coverage for academic details and decision outcomes. country and intake appear to be key segmentation variables for understanding applicant distribution. A major next step for later weeks will be deciding which high-missingness columns should be dropped, imputed, or treated as optional fields, depending on the analysis goal.

In [65]:
```
# 9) simple insights helpers
print("\nquick insight checks")
if "Country" in df.columns:
    print("\ntop 10 countries")
    print(df["Country"].value_counts().head(10))

if "Intake" in df.columns:
```

```
    print("\nintake distribution")
    print(df["Intake"].value_counts(dropna=False).head(15))

print("\ncolumns with over 80 percent missing")
print(missing_pct[missing_pct > 80].index.tolist())
```

```
quick insight checks

top 10 countries
Country
India            4617
United states     764
Ghana             440
Nigeria           430
Pakistan          273
China             125
Nepal              81
Iran               75
Canada             69
Kenya              62
Name: count, dtype: int64

intake distribution
Intake
Computer Science - MS: 2024 Fall (September '24)              504
Business Analytics - MS: 2024 Fall (September '24)            419
Data Science - MS: 2024 Fall (September '24)                  254
Cybersecurity - MS: 2024 Fall (September '24)                 181
Business Analytics - MS: 2025 Fall (September '25)            160
Artificial Intelligence - MS: 2024 Fall (September '24)       160
Health Informatics - MS: 2024 Fall (September '24)            155
Business Analytics - MS: 2025 Winter (January '25)            144
Business Analytics - MS: 2024 Winter (January '24)            143
Human-Computer Interaction - MS: 2024 Fall (September '24)    140
Information Systems - MS: 2024 Fall (September '24)           138
Finance - MS: 2024 Fall (September '24)                       127
Computer Science - MS: 2025 Winter (January '25)              126
Computer Science - MS: 2024 Winter (January '24)              122
Supply Chain Management - MS: 2024 Fall (September '24)       116
Name: count, dtype: int64

columns with over 80 percent missing
['Most_Recent_Released_Decision', 'SLU_Start_Comment', 'RIT_Email_Created', 'Housing
_Contract', 'Is_Global_Grad', 'Is_Admitted', 'SEVIS_ID', 'Official_University_email_
address', 'Application_Agency_Code', 'I_901_Status', 'SLU_Start_City', 'Outstanding_
Checklist_Items', 'Date_of_Birth', 'Prior_I-20_Outreach_Detail', 'Prior_Non_I-20_Out
reach_Detail', 'Most_Recent_Contact', 'Most_Recent_User_and_Date', 'Recent_GT_Form_I
nitiative', 'Escalation_Required', 'Final_Result', 'Caller_Name_2', 'Date_of_Contact
_2', 'Phone_Number_2', 'Outcome_2', 'City_and_Branch', 'Status', 'Category', 'Major_
1st_Choice', 'Header', 'Counsler', 'Status-2', 'Template', 'Street_3', 'Degree_Typ
e', 'Major', 'Intake-2', 'Region-2', 'Citizenship', 'Remark_2', 'Remark']
```

In [64]:

```
df.to_csv("depaul_data_cleaned_for_eda.csv", index=False)
print("\neda complete, notebook ran successfully")
```

```
eda complete, notebook ran successfully
```

In [ ]:

```

```